John Dave C. Pega

Software Developer

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ABOUT

A normal guy who knows how to type — and build systems. Versatile developer with hands-on experience across embedded systems, machine learning, game development, full-stack web, and language design. Open to opportunities.

SKILLS

Python · C++ · JavaScript · TypeScript · Java · C# · C · Bash · Dart · Flask · REST APIs · Sequelize · Socket.IO · Raspberry Pi · Embedded Design · Sensor Integration · Event-Driven Systems · Multithreading · Image Processing · Q-Learning · NEAT · Data Pipelines · Complex Game Dev · Game Physics · Language Design · Compiler Basics · Android Dev · Automation

EDUCATION

City College of Calamba
Bachelor of Science in Computer Science
2022 – 2026 GPA: 3.6/4.0

CERTIFICATIONS

Wadhwani Employability Skills Certificate 2026
Wadhwani Foundation
https://wadhwanifoundation.org

LANGUAGES

English — Intermediate
Filipino — Native

HOBBIES

Reading · Open Source · Programming · Chess · Automation

EXPERIENCE

Software Developer 2022 - Present
Freelance

- Worked directly with clients to gather requirements and deliver tailored solutions.
- Maintained and optimized existing projects for performance and usability.
- Developed 8 academic thesis projects for clients, including system design, documentation, and implementation.
- Built 7 organizational websites, focusing on responsive design, performance, and user experience.
- Delivered end-to-end solutions from requirements gathering to deployment and maintenance.

PROJECTS

AGRI-BOT: AI-Driven Robotics System for NFT Hydroponic Plant Disinfection

Raspberry Pi · Python · YOLOv8 · Mask R-CNN · Node.js (Express) · React · Electron · Flutter · Socket.IO · VPN Networking

Designed and developed a full-stack AI-powered robotics system for automated vegetable plant disinfection in NFT hydroponic environments. The system integrates a Raspberry Pi-based robot controller with computer vision models (YOLOv8, Mask R-CNN) for detection and analysis, scheduling, and operational automation algorithms. Built a multi-platform control and monitoring system including web studio, desktop application (Electron), and mobile app (Flutter). Implements real-time communication via Socket.IO and secure remote access through VPN-based authentication tied to unique robot IDs, enabling global monitoring through a 24/7 live camera feed.

JDM Programming Language

https://github.com/JDM-Github/jdm-language

C/C++

Built a programming language from scratch with no external libraries except SDL2. Implemented core language design, execution system, and developed a custom GUI framework allowing simplified creation of graphical applications.

JDM-CLI Plugin-Based Developer Toolkit

https://github.com/JDM-Github/jdm-cli

Node.js / CLI Architecture

Developed a modular command-line interface system supporting plugin-based architecture. Users can install, manage, and run plugins such as full-stack project generators. Includes ecosystem features like plugin installation, updates, removal, and execution (e.g., electron-flask scaffolding with React, Flask, and Electron integration).

PUBLICATIONS

AGRI-BOT: AI-DRIVEN ROBOT FOR VEGETABLE PLANT DISINFECTION IN NUTRIENT FILM TECHNIQUE (NFT) HYDROPONIC SYSTEM USING YOU ONLY LOOK ONCE (YOLOV8) AND MASK REGION- BASED CONVOLUTIONAL NEURAL NETWORK (MASK R-CNN) ALGORITHMS 2026
City College of Calamba